

Learning for Data Science

Course 2: Machine Learning for Data Science

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Université
de Montréal



Mila

CIFAR

Outline

Data science

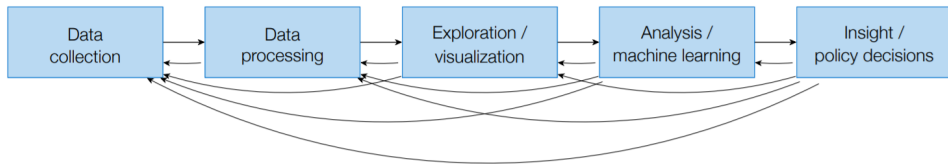
Categories of Machine Learning

Regression: applications

Classification: Applications

Unsupervised Learning — Why?

Data science



- Why do we want machine learning?
- Source: <https://www.datasciencecourse.org/slides/intro.pdf>

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 - Reinforcement Learning (Relative comparison between data),
 - semi-supervised learning (Only a portion of output available for each x_i),
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- Sometime, unclear... Q: what does supervision mean?

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- Person's characteristics $\rightarrow$ Whether they'll vote
- Marketing plan $\rightarrow$ Total sales amount
- Image pixel values $\rightarrow$ cat or dog



$x_i = (x_i^1, \dots, x_i^m) \leftarrow$ all the inputs

$\hat{y}_i \leftarrow f(x_i|\theta) \leftarrow$ inputs to output relationship

Learn a function linking input and output

- f is a “simple” function that describes the relationship between input x and output $\hat{y}$, parameterized by θ :

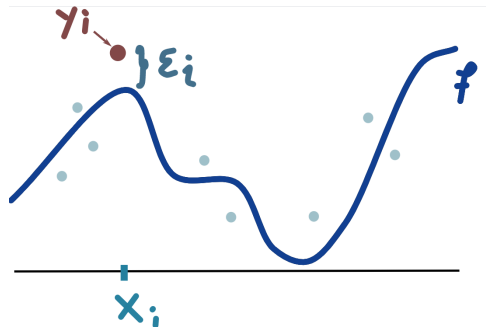
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- ϵ_i This represents the error, the difference between model output and a truth value.



Supervised Learning: Input - Output

Old label: pier
Real: dock; pier;
speedboat; sandbar;
seashore



Old label: hammer
Real: screwdriver;
hammer; power drill;
carpenter's kit



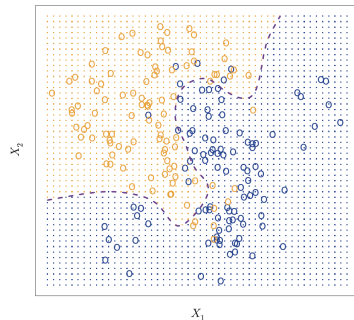
the	department,
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- Goal: Given an input x , predict an output y
- What we have: Observations:
- Why it is “hard”:
- Labels may be stochastic:
- The prediction may be “complex” (non-linear in high dimension)
- Only few pairs observed. (Bad labels)
- Formally: $(x_i, y_i), i = 1, \dots, n$

Supervised Learning: Classification

- Prediction of discrete classes.
(e.g. cats vs dogs)
- Example here:
 - Input: 2D position
 - Output (Labels) blue or yellow
- Goal: each region of the space should be blue or yellow.
- Evaluation: 0-1 loss on test set



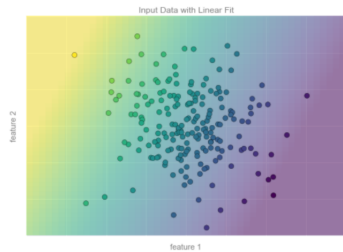
Supervised Learning: Regression

- Prediction of continuous labels.
(e.g. temperature)
- Example here:
 - Input: 2D position
 - Output: value (represented by a color)
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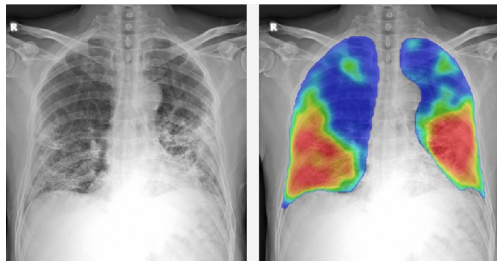
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- What if our data contained a y of value 7.8?

Help with medical diagnosis

- Aim: To predict the probability of the presence of Covid-19 in the lungs of patients based on the following factors:
- X-rays
- Laboratory tests
- Medical file
- Way of life

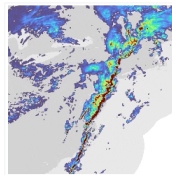


- COVID-19 catch-up: New AI tool detects affected lung tissue by analyzing X-Ray images | Silicon Channels

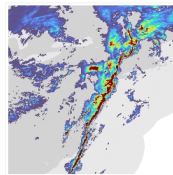
Weather prediction

- Goals: Predict global rainfall from
- Satellite images
- Buoy surveys
- Historical observations
- El Niño and La Niña years
- etc

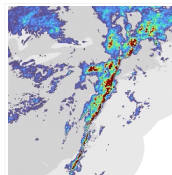
- Source:
<https://www.deepmind.com/blog/now-the-next-hour-of-rain>



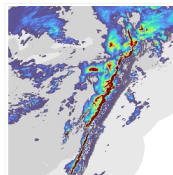
Target



DGMR



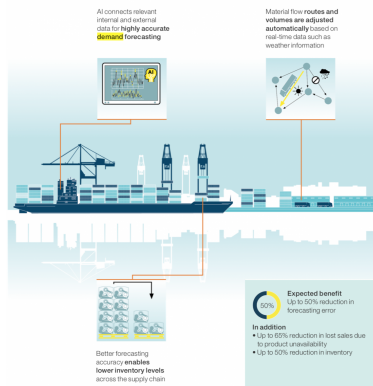
PySTEPS



UNet

Supply and demand predictions

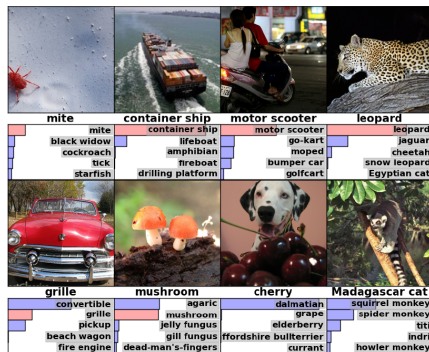
- Goals: To foresee the following aspects for an international trading company
- Inventory evolution
- Evolution of demand



- True Stock | Top 5 Benefits of using Machine Learning for Demand Forecasting

Object recognition

- Aim: to recognize the types of objects present in an image despite the variations of objects and poses.



- A. Krizhevsky, I. Sutskever and G.E. Hinton. Imagenet Classification with Deep Convolutional Neural Networks. In Advances in Neural Information Processing Systems, 2012.

Recognition of handwritten characters

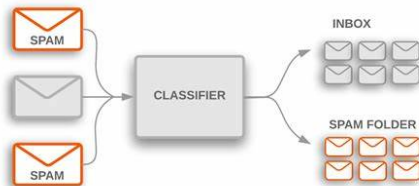
- Aim: to recognize each character despite different writing styles.



- By Josef Steppan, CC BY-SA-4.0.
<https://commons.wikimedia.org/wiki/File:Mni>

Natural language processing

- Purpose: to recognize combinations and arrangements of words to filter SPAM/SPAM.



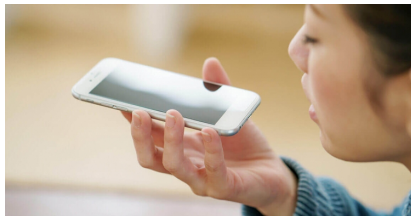
- Decision Trees

Speech Recognition

- Goals: identify a speaker from
- Time dependence of information.
- Dictionaries of valid words/structures.



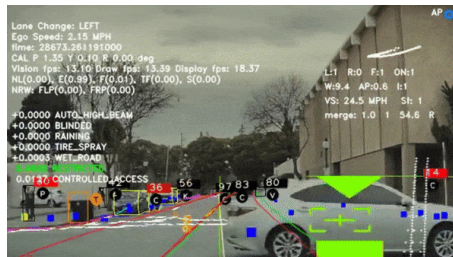
- ABC



- Siri richtig nutzen: nützliche Tipps & Befehle auf einen Blick (sparhandy.de)

Autonomous car

- Purpose:
- to identify the pixels corresponding to the
- pedestrians
- buildings
- trees
- types of terrain
- despite the movements of the objects and the car!
- We are interested in areas of the image containing objects of the same type (semantics).

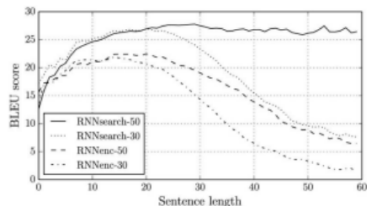


Very important to have robust systems

- Robust to different situations.
- Especially the ones that have never been experienced before.
- Humans are very good with variability.

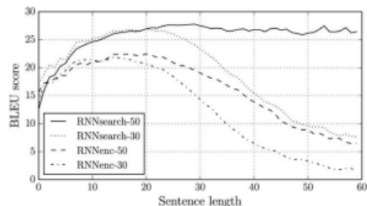


Next Token Prediction (LLMs)



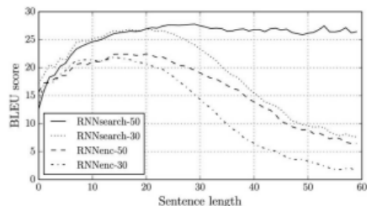
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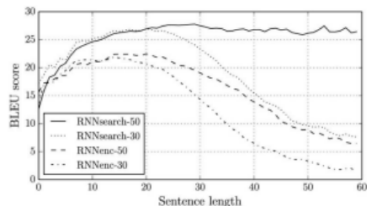
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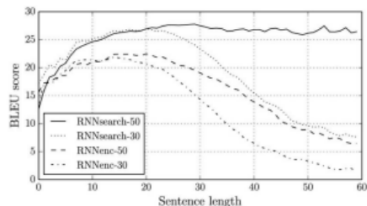
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- Object Detection (ex: FD-SwinV2-G Transformer)

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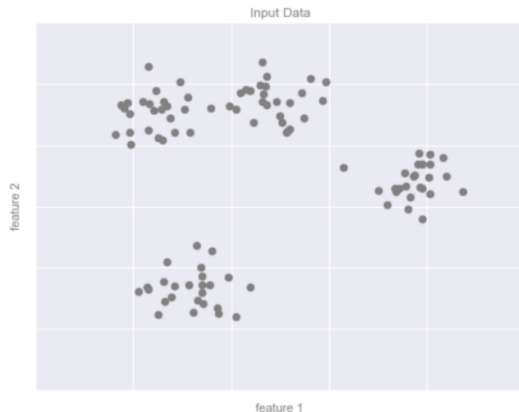
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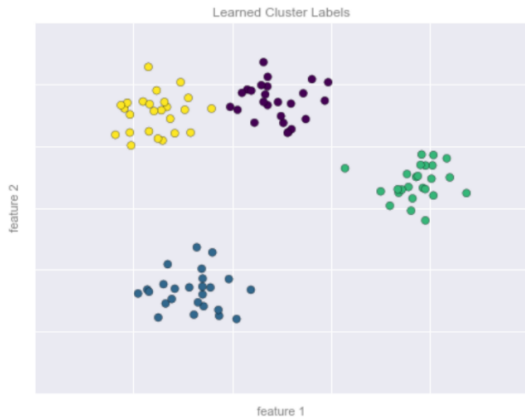
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- Example: data points close with respect to a certain distance (which distance???)
- Importance of the right features: will better capture similarity.

Clustering

- By eye it is clear that there are 4 clusters!
- Importance of data visualization!!!!(now we know that we need to look for 4 clusters)



Clustering



Dimensionality reduction: Inferring structure of unlabeled data

- Problem: Data can be very high dimensional, for example images.
- Hypothesis: There exists a low-dimensional manifold latent z that captures the data sufficiently.
 - This is known as the manifold hypothesis. That in many cases there's a lower dimensional space that still captures the data. (Cayton and others, 2005)

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- function: $\hat{z}_i \leftarrow f(x_i|\theta)$
 - $\hat{z}_i$ is often continuous



Dimensionality reduction

- Why this hypothesis?
- Uniformly Random generated examples do not look like *real* data.



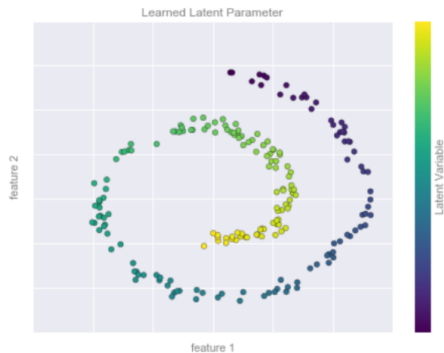
Illustration of the manifold hypothesis

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- Example: Principal Component Analysis (covered in week 7)

Dimensionality reduction: conclusion

- Maybe the less clear setting
- High level idea: find better representations
- Example: Principal Component Analysis (covered in week 7)
- Useful to eventually solve other tasks (e.g. classification with limited data)
 - ML using few features

Conclusion (part 1)

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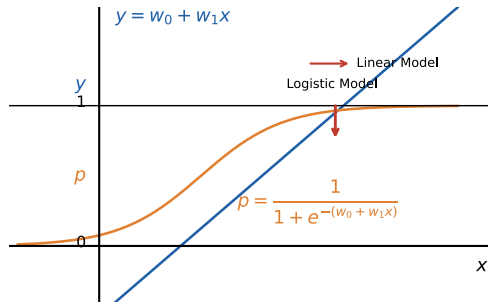
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- How to use these algorithms in practice (with sk-learn) in future labs.

Logistics Regression Detailed Example

- Logistic Regression:
 - Estimating $P(C_1|\mathbf{x})$
- Learn w and w_0 from $\mathcal{X} = \{\mathbf{x}_i, y_i\}$,
 - with the labels $y_i \in \{0, 1\}$.
 - $\theta = w_1, w_0$

$$y = \hat{P}(C_1|\mathbf{x}) = \frac{1}{1 + \exp[-(\mathbf{w}^\top \mathbf{x} + w_0)]}$$

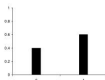


Logistics Regression

Discrete distributions

The Bernoulli distribution

$$p(x) = P[X = x] = \begin{cases} q = 1 - p & x = 0 \\ p & x = 1 \end{cases}$$



$$X = \begin{cases} 1 & \text{Bernoulli trial} = \mathbf{S} \\ 0 & \text{Bernoulli trial} = \mathbf{F} \end{cases}$$

- y_i for some $\mathbf{x}_i$ follows a Bernoulli distribution with probability $\hat{y}_i = P(C_1|\mathbf{x}_i)$

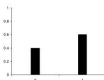
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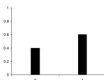
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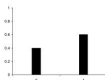
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- Error to maximize log-likelihood
- Also called cross-entropy.

$$l(\mathbf{w}, w_0|\mathcal{X}) = \prod_i (\hat{y}_i)^{y_i} (1 - \hat{y}_i)^{(1-y_i)}$$

$$E_{entr}(\mathbf{w}, w_0|\mathcal{X}) = -\log l(\mathbf{w}, w_0|\mathcal{X}) = -\sum_i y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)$$

Learning algorithm

- Randomly initialize the weights, $w^j \sim U(-0.01, 0.01)$.
- Repeat until convergence:
- Prediction:

$$\hat{y}_i = \frac{1}{1 + \exp[-(\mathbf{w}^\top \mathbf{x}_i + w_0)]}, \quad i = 1, \dots, N(\text{batch})$$

- Gradient step

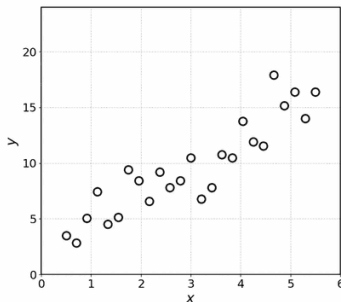
$$w^j = w^j + \eta \sum_i (y_i - \hat{y}_i) x_i^j, \quad j = 1, \dots, D(\text{each feature})$$

$$w_0 = w_0 + \eta \sum_i (y_i - \hat{y}_i)$$

Gradient descent (video)



1. Intuition



This animation is embedded directly in the PDF as a sequence of frames (Acrobat/Adobe Reader) — no external video file needed.

Interpretation

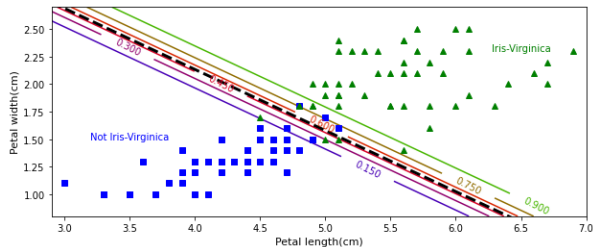
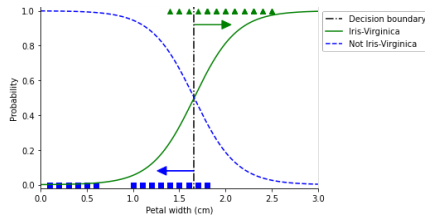
- Prediction:

$$\hat{y} = \hat{P}(C_1|\mathbf{x}) = \frac{1}{1 + \exp[-(\mathbf{w}^\top \mathbf{x} + w_0)]}$$

- Interpretation of w^j coefficients:
 - the larger w^j , the more important x^j is for the prediction.
 - If w^j is zero, feature x^j does not affect the prediction.

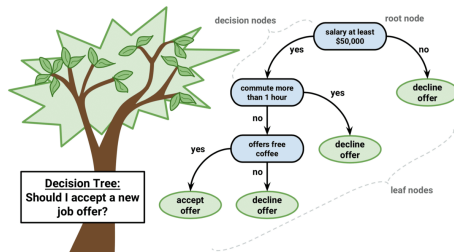
Example on the IRIS dataset

- 4 features: the length and width (in centimeters) of the sepals and petals of the flowers.
- Proposed by Fisher in 1936.
- Two of the three species were collected in Gaspésie (Quebec, Canada).



Decision tree

- Performs a hierarchical splitting of the input space
- Each node corresponds to a test on a characteristic (input variable)
- The leaves of the tree correspond to the predictions



Characteristics of Decision Trees

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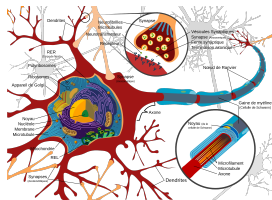
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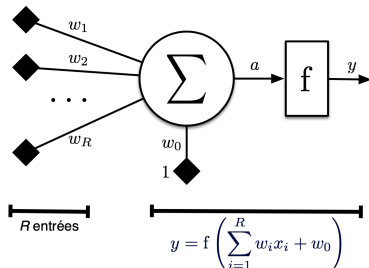
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- Classifiers with low bias and high variance leads to Instability
- (concept of bias and variance will be seen in more detail in future courses.)

Neural Networks

Artificial neuron



- Inspired by biological neurons
- A neuron is an affine function whose output is subject to an activation function
- Linear discriminant

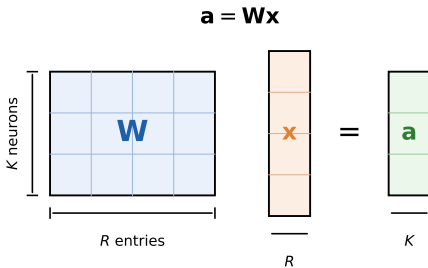
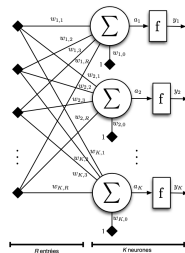


$f(a) = \frac{1}{1+e^{-a}} \Rightarrow y = \hat{P}(C_1|\mathbf{x})$: our logistic regression!

Layer

- A layer is a set of vertically stacked neurons
- Each neuron processes input and produces an output
- Linear discriminant

$\mathbf{y} = f(\mathbf{W}\mathbf{x} + \mathbf{w}_0)$, with $\mathbf{W} \in \mathbb{R}^{K \times R}$ now a matrix of weights (one row per neuron)



Multilayer Perceptron

- Several connected layers form a deep artificial neural network

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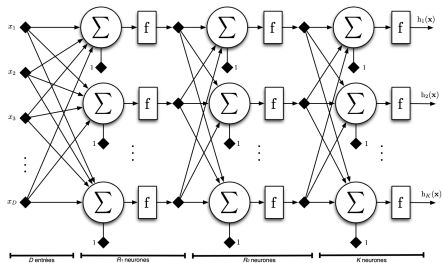
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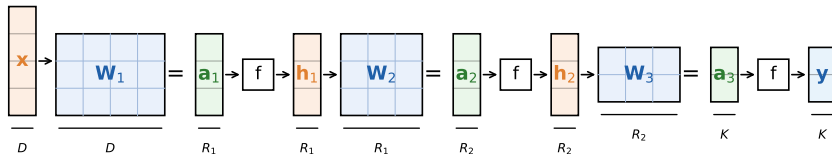
Multilayer Perceptron

- Several connected layers form a deep artificial neural network
- Three types of layers: input, output, hidden layers
- The outputs of one layer are the inputs of the next
- Discriminant linear or not according to the activation function

Multilayer Perceptron



$$\mathbf{y} = f(\mathbf{W}_3 f(\mathbf{W}_2 f(\mathbf{W}_1 \mathbf{x})))$$



Activation function

- Function applied to the output of an artificial neuron
- There are several functions whose usefulness depends on the context:
 - Relu:

$$f(x) = \max(x, 0)$$

- Sigmoid:

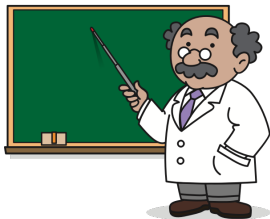
$$f(x) = \frac{1}{1 + e^{-x}}, \quad x \in [-\infty, \infty], \quad f(x) \in [0, 1]$$

- Hyperbolic Tangent:

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad x \in [-\infty, \infty], \quad f(x) \in [-1, 1]$$

Last layer

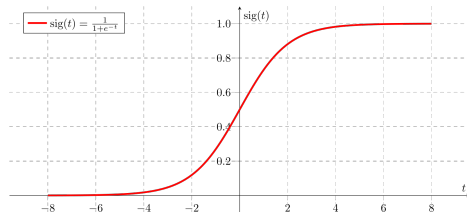
- The activation function of the last layer of an artificial neural network allows to adapt the output to the task we want to accomplish



Example: binary classification

- The Sigmoid function transforms an output into a probability between 0 and 1
- Useful for binary classification
- Output compared to 0.5 (50%)

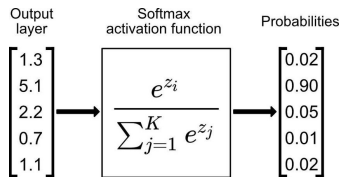
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Example: multiclass classification

- The Softmax function transforms a vector of outputs into a discrete probability distribution

$$f(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

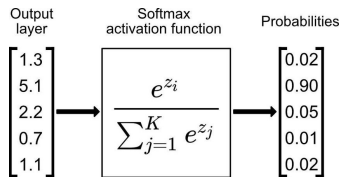


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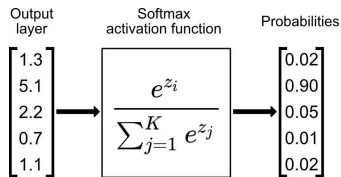


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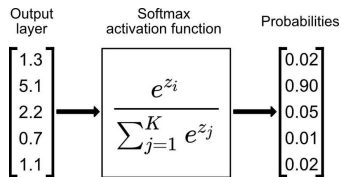
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- Indeed, we have $f(\mathbf{z})_i \in [0, 1]$ and $\sum_{i=1}^K f(\mathbf{z})_i = 1$
- Very useful for multiclass classification
- The class with the highest probability is chosen at test time

$$\hat{y} = \arg \max_i f(\mathbf{z})_i \leftarrow \text{class chosen at test time}$$

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- The identity function used as an activation for the hidden layers serves as a linear discriminant

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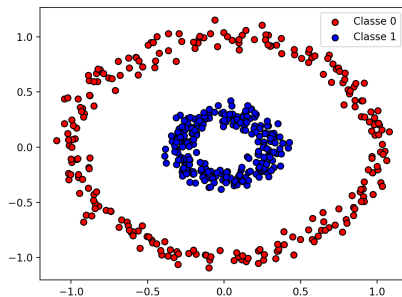
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- Nonlinearity allows networks to learn
- Without nonlinearity, a network is a linear filter



Example: Data classification with a perceptron

- Distribution of data to be classified
 - 2-D data
 - Two well-separated classes
 - Non-linearly separable boundary

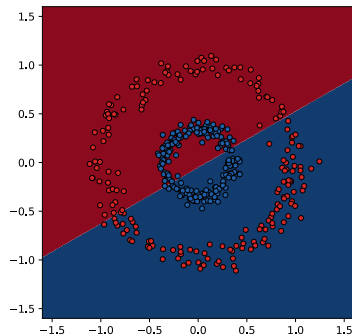


Example: linear network

Try it yourself in Google Colab

- A hidden layer: 4 neurons
- Hidden layer activation: identity
- The network poorly separates classes.

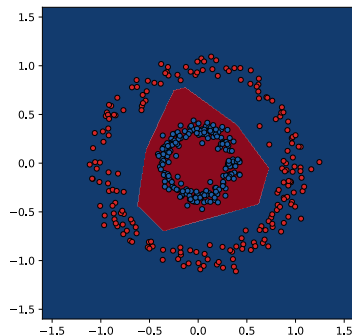
$$f(x) = x$$



Example: non-linear network

Try it yourself in [Google Colab](#)

- A hidden layer: 4 neurons
- Hidden layer activation: ReLU
- The non-linearity of ReLU allows the network to separate classes well.



$$f(x) = \max(0, x)$$

What learning method should be used?

- Log in with your first name, and your student number as your last name.
 - (A) Classification (B) Regression (C) Clustering (D) Dimensionality reduction
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- We have cat and dog images, but only want the ordinal (0,1) in a table.
- Sometimes the solutions depends on the data, other times on the desired outcome

Refs used to build this course

- Zico Kolter's course on Data Science.
- MOOC IVADO
- ([Witten *et al.*, 2016](#))

- Lawrence Cayton et al. Algorithms for manifold learning. *Univ. of California at San Diego Tech. Rep*, 12(1-17):1, 2005.
- Ailsa H Land and Alison G Doig. An automatic method for solving discrete programming problems. In *50 Years of Integer Programming 1958-2008: From the Early Years to the State-of-the-Art*, pages 105–132. Springer, 2009.
- Ian H. Witten, Eibe Frank, Mark A. Hall, and Christopher J. Pal. *Data Mining: Practical Machine Learning Tools and Techniques*. Morgan Kaufmann, Amsterdam, 4th edition, 2016.